

Size Knowledge Accelerates One-Bit Las Vegas Broadcast in Anonymous Dynamic Networks

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Abstract

We study the running-time value of network-size knowledge for a two-state private-coin protocol for idle-start stabilizing broadcast in anonymous, port-indistinguishable, synchronous, 1-interval-connected dynamic networks. For Bernoulli Reactivation Flooding (BRF), an active node broadcasts and survives independently with probability p , while an idle node becomes active after receiving the broadcast. Fair-coin BRF requires no knowledge of n but has adaptive worst-case expected stabilization time $2^{\Theta(n^2)}$. We show that when n is known, the same two persistent protocol states suffice for expected time $2^{\Theta(n \log n)}$: it is enough to choose a size-aware drop probability $\delta_n = 2^{-k_n}$, where $k_n = \lceil \log_2((n+1)/2) \rceil$. The bias is implementable from fair private coins without additional persistent protocol state. We also prove that for every fixed survival probability $p \in (0, 1)$, a causal state-adaptive topology adversary maximizes the Bellman continuation value by exposing exactly one idle frontier node whenever the active set is a proper subset of the network. Finally, we prove that $2^{\Theta(n \log n)}$ is asymptotically optimal over the homogeneous fixed-bias BRF family against causal state-adaptive adversaries. The optimality statement is only for this BRF family; it is not a lower bound for all two-state randomized broadcast protocols.

Keywords. anonymous dynamic networks; randomized broadcast; stabilizing termination; local memory; size knowledge; Bellman equations; adversarial dynamic networks.

1 Introduction

Broadcast with very small persistent local memory is sharply constrained by anonymity and adversarial topology changes. In anonymous synchronous 1-interval-connected dynamic networks, Parzych and Daymude proved that deterministic idle-start broadcast with stabilizing termination requires superconstant local memory, while their Countdown algorithm uses $O(\log n)$ memory [1]. Turau subsequently obtained a randomized $O(\log \log n)$ -space algorithm with high-probability guarantees when n is known, but in the non-idle-start regime [2]. These results leave open how far private randomness can compress persistent state under the stricter idle-start requirement, and what global knowledge can buy once the state budget has already collapsed to a constant.

Our direct predecessor introduced Bernoulli Reactivation Flooding (BRF), a two-state idle-start randomized protocol with zero error, almost-sure stabilizing termination, and finite expected

stabilization time for every fixed bias $p \in (0, 1)$ [5]. For the fair-coin choice $p = 1/2$, the causal state-adaptive worst-case expectation is $2^{\Theta(n^2)}$. Thus randomness removes the deterministic idle-start memory barrier in this protocol, but the most elementary bias pays an enormous time price.

This paper asks a narrower question: *if the network size n is known, can that knowledge accelerate the same two-state protocol without adding persistent protocol state?* The answer is yes. Choosing a drop probability on the order of $1/n$ reduces the worst-case expected time from the fair-coin $2^{\Theta(n^2)}$ scale to $2^{\Theta(n \log n)}$. Moreover, this order is optimal within the entire homogeneous fixed-bias BRF family against causal state-adaptive topology adversaries.

Contributions.

1. We give a dyadic size-aware BRF bias $\delta_n = 2^{-k_n}$ with $k_n = \lceil \log_2((n+1)/2) \rceil$. It keeps exactly two persistent protocol states and satisfies

$$2^{nk_n} \leq W_n(p_n) \leq ne^n 2^{nk_n},$$

hence $W_n(p_n) = 2^{\Theta(n \log n)}$.

2. For every fixed $p \in (0, 1)$, we prove that the causal state-adaptive adversary maximizes the Bellman continuation value by exposing exactly one idle frontier node whenever the active set is a proper subset of the network.
3. We prove the fixed-bias minimax law

$$\inf_{p \in (0, 1)} W_n(p) = 2^{\Theta(n \log n)}.$$

This is a family-specific result, not a lower bound for all two-state randomized broadcast protocols.

4. We provide reproducible exact-rational Bellman evaluations for rational biases and high-precision numerical bias optimization for $2 \leq n \leq 20$. The experiments reproduce the predecessor's exact fair-coin values and show that the finite-size runtime optimum lies very close to the bias maximizing the uniform block-success certificate.

2 Related work and scope

Parzych and Daymude study anonymous, synchronous, 1-interval-connected dynamic broadcast without identifiers or port labels. For idle-start stabilizing termination they prove a deterministic $\omega(1)$ memory lower bound and give the $O(\log n)$ -space Countdown algorithm [1]. Our result does not contradict that lower bound because BRF is randomized; our state claim is an upper bound for persistent protocol control state under private randomness.

Turau studies randomized broadcast in the same broad anonymous dynamic setting and obtains an $O(\log \log n)$ -space, constant-message algorithm with high-probability stabilization when n is known [2]. That algorithm is explicitly non-idle-start: uninformed nodes may transmit before receiving the broadcast. The present paper instead keeps the idle-start restriction and asks only how size knowledge changes the running time of a particular two-state Las Vegas protocol family.

Randomized flooding also appears in static port-aware models. Austin, Gadouleau, Mertziou, and Trehan define Random Flooding as a randomized relaxation of Amnesiac Flooding [3]. Their nodes act per neighbor in a fixed graph, whereas our model is port-indistinguishable local broadcast

on an adversarially changing connected topology. We therefore make no generic novelty claim for randomized flooding as a mechanism.

Blanc, Di Luna, and Viglietta study deterministic computation in anonymous dynamic networks with one-bit *messages* [4]. Their one-bit resource is communication width, while the resource counted here is persistent local protocol state. These notions should not be conflated.

Finally, the fair-coin BRF analysis and its $2^{\Theta(n^2)}$ causal state-adaptive worst-case runtime come from the direct predecessor [5]. The new claims here are the size-aware runtime improvement, arbitrary-bias one-frontier Bellman optimality, and the fixed-bias BRF-family minimax characterization.

3 Model and protocol family

We use the anonymous, synchronous, port-indistinguishable local-broadcast model on a fixed node set V , $|V| = n \geq 2$, with a connected communication graph in every round. There is a unique broadcaster. The task is idle-start broadcast with stabilizing termination, not termination detection.

A node has one persistent protocol bit, with states **ACTIVE** and **IDLE**. Initially only the broadcaster is active. In every round, every active node broadcasts **INF**. An active node independently remains active with probability p and becomes idle with probability

$$\delta := 1 - p.$$

An idle node becomes active in the next round iff it received at least one copy of **INF** in the current round. Active nodes do not renew their active state in response to received copies of **INF**.

We count only persistent protocol control state. The known value n is a read-only problem parameter and is not charged to the protocol-state count. The theorems below are stated directly for a randomized transition kernel with survival probability p . If the local random source is restricted to fair private coins and atomic local computation is free, the dyadic bias used below can equivalently be sampled from a finite prefix of the private coin tape within one round. No such transient random bits are retained across rounds.

Lemma 1 (No false extinction). *Let S_t be the active set at the start of round t . If $\emptyset \neq S_t \neq V$, then $S_{t+1} \neq \emptyset$ deterministically. Hence a nonempty proper active set cannot jump directly to $\emptyset$. In particular, on the first transition into the absorbing empty state, the preceding active set must be all of V ; therefore every node has already been informed before absorption.*

Proof. If S_t is a nonempty proper subset of V , connectivity of the current snapshot gives an edge from an active node to an idle node. The active endpoint broadcasts **INF**, so the idle endpoint is active in the next round regardless of all survival coins. Thus a nonempty proper active set cannot become empty. Now consider the first time $t + 1$ for which $S_{t+1} = \emptyset$. By firstness, $S_t \neq \emptyset$; the preceding argument excludes S_t being a proper subset of V , so $S_t = V$. Once the active set is empty, nobody broadcasts and no idle node can reactivate, so the empty state is absorbing. $\square$

For a fixed n and fixed bias p , let $W_n(p)$ denote the supremum of the expected stabilization time over causal state-adaptive 1-interval-connectivity adversaries.

4 Size-aware dyadic bias

Define

$$k_n := \left\lceil \log_2 \frac{n+1}{2} \right\rceil, \quad \delta_n := 2^{-k_n}, \quad p_n := 1 - \delta_n.$$

Then

$$\frac{1}{n+1} < \delta_n \leq \frac{2}{n+1}.$$

Under the fair-coin realization just described, an active node becomes idle iff its first k_n fair coin flips are all zero. The flips may stop at the first one, so the expected number of fair random bits consumed by an active update is

$$\sum_{i=1}^{k_n} 2^{-(i-1)} < 2,$$

while the worst-case number is $k_n = O(\log n)$. This sampling procedure does not add persistent protocol states.

Lemma 2 (Uniform block success). *For every $p \in (0, 1)$, every execution history ending in a nonempty active set, and every causal continuation of connected snapshots, BRF reaches the absorbing all-idle state within the next n rounds with conditional probability at least*

$$q_n(p) := p^{n(n-1)/2}(1-p)^n.$$

Consequently,

$$W_n(p) \leq \frac{n}{q_n(p)}.$$

Proof. Consider the event that until the active set first becomes all of V , every currently active node survives, and then all n active nodes drop simultaneously. While the active set is a nonempty proper subset, connectivity forces at least one idle node to receive INF. On the chosen event the active set therefore grows strictly until it reaches V , in at most $n-1$ rounds. The total number of required survival outcomes is at most $1 + \dots + (n-1) = n(n-1)/2$, followed by n simultaneous drop outcomes. Thus the event has conditional probability at least $q_n(p)$ and finishes within n rounds. Repeating the argument in n -round blocks gives a geometric tail bound and the displayed expectation bound. $\square$

Lemma 3 (Universal simultaneous-drop lower bound). *For every $p \in (0, 1)$, every causal topology adversary, and $\delta = 1 - p$,*

$$\mathbb{E}[T] \geq \delta^{-n}.$$

Proof. The first transition into the absorbing empty state can occur only from a round that starts with all n nodes active, and then only if all n nodes independently drop. Hence, conditioned on any nonabsorbed history, the probability of absorption in the next round is at most δ^n . Inductively,

$$\mathbb{P}[T > t] \geq (1 - \delta^n)^t.$$

Summing the tail yields

$$\mathbb{E}[T] \geq \sum_{t \geq 0} (1 - \delta^n)^t = \delta^{-n}.$$

$\square$

Theorem 4 (Known-size one-bit runtime). *For the dyadic size-aware bias above,*

$$2^{nk_n} \leq W_n(p_n) \leq ne^{nk_n}.$$

Therefore

$$\log_2 W_n(p_n) = n \log_2 n + O(n),$$

and hence

$$W_n(p_n) = 2^{\Theta(n \log n)}.$$

Proof. The lower bound follows from the previous lemma because $\delta_n^{-n} = 2^{nk_n}$.

For the upper bound,

$$W_n(p_n) \leq n p_n^{-n(n-1)/2} \delta_n^{-n}.$$

Since $\delta_n \leq 2/(n+1)$,

$$-\log p_n = -\log(1 - \delta_n) \leq \frac{\delta_n}{1 - \delta_n} \leq \frac{2}{n-1}.$$

Therefore

$$p_n^{-n(n-1)/2} \leq e^n,$$

which gives

$$W_n(p_n) \leq n e^n 2^{nk_n}.$$

Finally, $k_n = \log_2 n + O(1)$. □

Certificate-optimal continuous bias. The uniform n -round success certificate in Lemma 2 is

$$q_n(p) = p^{a_n} (1-p)^n, \quad a_n := \frac{n(n-1)}{2}.$$

Maximizing $\log q_n(p) = a_n \log p + n \log(1-p)$ gives

$$\frac{a_n}{p} - \frac{n}{1-p} = 0,$$

and therefore the unique certificate maximizer is

$$p_n^* = \frac{n-1}{n+1}, \quad \delta_n^* = \frac{2}{n+1}.$$

This optimizes only the uniform block-success certificate, not the actual expected stabilization time. The dyadic choice satisfies

$$\frac{\delta_n^*}{2} < \delta_n \leq \delta_n^*,$$

so it is a fair-coin-implementable discretization within a factor two in drop probability.

5 General-bias adaptive Bellman structure

Suppose the current active-set size is $k < n$. A state-adaptive connected topology can expose exactly b idle nodes to the active set, where $1 \leq b \leq n - k$. Those b idle nodes are deterministically active next round, while each of the k current active nodes independently survives with probability p . Thus

$$K_{t+1} = b + \text{Bin}(k, p).$$

Consider the one-frontier policy $b = 1$, and let $\hat{E}_k$ be its expected remaining stabilization time from k active nodes. Define

$$\Delta_k := \hat{E}_k - \hat{E}_{k+1}, \quad 1 \leq k < n.$$

Lemma 5 (General-bias difference recurrence). *For every $1 \leq k < n$,*

$$p^k \Delta_k = 1 + \sum_{r=1}^{k-1} \Delta_r \mathbb{P}[\text{Bin}(k, p) \leq r-1].$$

In particular,

$$\Delta_k > 0$$

for every $1 \leq k < n$.

Proof. Under one-frontier exposure,

$$\widehat{E}_k = 1 + \sum_{j=0}^k \binom{k}{j} p^j (1-p)^{k-j} \widehat{E}_{j+1}.$$

For $0 \leq j \leq k$,

$$\widehat{E}_{j+1} = \widehat{E}_{k+1} + \sum_{r=j+1}^k \Delta_r.$$

Substitution gives

$$\Delta_k = 1 + \sum_{r=1}^k \Delta_r \mathbb{P}[\text{Bin}(k, p) \leq r-1].$$

The $r = k$ probability equals $1 - p^k$. Moving that term to the left gives

$$p^k \Delta_k = 1 + \sum_{r=1}^{k-1} \Delta_r \mathbb{P}[\text{Bin}(k, p) \leq r-1].$$

For $k = 1$, this gives $\Delta_1 = p^{-1} > 0$. Induction now yields $\Delta_k > 0$ for every $k < n$. $\square$

Theorem 6 (One-frontier Bellman optimality for arbitrary bias). *For every fixed $p \in (0, 1)$, among all causal state-adaptive choices of frontier size b at a nonterminal state $k < n$, the Bellman-optimal choice is*

$$b = 1.$$

Proof. The previous lemma gives

$$\widehat{E}_1 > \widehat{E}_2 > \dots > \widehat{E}_n.$$

For any realization j of $\text{Bin}(k, p)$ and any feasible $b \geq 1$,

$$\widehat{E}_{1+j} \geq \widehat{E}_{b+j}.$$

Hence, for every nonterminal state $k < n$,

$$\widehat{E}_k = 1 + \mathbb{E}[\widehat{E}_{1+\text{Bin}(k,p)}] \geq 1 + \mathbb{E}[\widehat{E}_{b+\text{Bin}(k,p)}].$$

At $k = n$ the topology has no effect on the next active-set size, so the same value function satisfies the unique boundary transition there, with $\widehat{E}_0 = 0$.

This is a truncation-based verification argument. By the uniform block-success lemma, every admissible policy has finite expected absorption time; in particular the one-frontier values $\widehat{E}_j$ are finite for all j . Fix any admissible causal state-adaptive policy π and let T be its absorption time. Conditional on the history at time $t < T$, apply the displayed inequality to the policy's current feasible choice of b . Equivalently,

$$1 + \mathbb{E}_\pi[\widehat{E}_{K_{t+1}} \mid \mathcal{F}_t] \leq \widehat{E}_{K_t}.$$

Thus the stopped process

$$Z_t := \widehat{E}_{K_{t \wedge T}} + (t \wedge T)$$

is an integrable supermartingale. For the bounded stopping time $T \wedge m$, optional stopping gives

$$\mathbb{E}_\pi[T \wedge m] + \mathbb{E}_\pi[\widehat{E}_{K_{T \wedge m}}] \leq \widehat{E}_k.$$

Because $\widehat{E}_0 = 0$ and $\widehat{E}_j \geq 0$, this implies $\mathbb{E}_\pi[T \wedge m] \leq \widehat{E}_k$ for every m . Monotone convergence then yields $\mathbb{E}_\pi[T] \leq \widehat{E}_k$. The one-frontier policy attains equality in the Bellman relation at every nonterminal state, so its value is exactly $\widehat{E}_k$ and it is Bellman-optimal. $\square$

6 Optimality within homogeneous fixed-bias BRF

We now prove only a family-specific result. It does not apply to arbitrary two-state randomized broadcast protocols.

Lemma 7 (Stopped coupling with the immigration–survival chain). *Fix $p \in (0, 1)$ and let the topology adversary use one-frontier exposure whenever the active set is a proper subset of V . Let*

$$\tau_n := \inf\{t : K_t = n\}.$$

Define an unrestricted auxiliary chain by

$$Y_0 = 1, \quad Y_{t+1} = 1 + \text{Bin}(Y_t, p),$$

using fresh independent survival coins.

There exists a coupling such that the stopped processes agree pathwise:

$$K_{t \wedge \tau_n} = Y_{t \wedge \sigma_n} \quad \text{for all } t,$$

where

$$\sigma_n := \inf\{t : Y_t = n\}.$$

In particular,

$$\tau_n = \sigma_n$$

under the coupling.

Moreover, for each fixed t ,

$$Y_t \stackrel{d}{=} 1 + \sum_{r=1}^t B_{t,r},$$

where $B_{t,1}, \dots, B_{t,t}$ are mutually independent and

$$B_{t,r} \sim \text{Bernoulli}(p^r).$$

(No joint independence across different values of t is asserted or needed.) Thus, with $X_t := Y_t - 1$,

$$\mu_t := \mathbb{E}[X_t] = \sum_{r=1}^t p^r < \frac{p}{1-p} < \frac{1}{1-p}.$$

Proof. Below n , the real one-frontier chain and the auxiliary chain have the same transition law,

$$k \mapsto 1 + \text{Bin}(k, p).$$

Couple them by using the same survival coin for each currently active individual and the same newly exposed frontier individual. If the two chains agree at a state $k < n$, they therefore agree at the next state. Also $1 + \text{Bin}(k, p) \leq k + 1 \leq n$, so the auxiliary chain cannot jump over n from a state below n . Hence the two chains agree pathwise until their common first hit of n .

For the distributional representation, regard the deterministic “1” in each update as one new immigrant. An immigrant born r rounds before time t is still present at time t iff it survives r independent Bernoulli- p updates, which has probability p^r . Distinct immigrants use disjoint survival coins, so the indicators are independent. Summing these indicators gives the claimed representation and expectation. $\square$

Lemma 8 (Slow all-active hitting when the drop rate is not too small). *Let $\delta = 1 - p$ and suppose*

$$\delta > n^{-1/2}.$$

For every $n \geq 3$ and every t ,

$$\mathbb{P}[Y_t \geq n] \leq \alpha_n := \left(\frac{e\sqrt{n}}{n-1} \right)^{n-1}.$$

Furthermore, for every $n \geq 14$,

$$\mathbb{E}[\tau_n] = \Omega(\alpha_n^{-1}) = 2^{\frac{1}{2}n \log_2 n - O(n)}.$$

Proof. By the coupling lemma,

$$X_t := Y_t - 1$$

is Poisson-binomial with

$$\mu_t < \frac{1}{\delta} < \sqrt{n}.$$

For $n \geq 3$, $n-1 > \sqrt{n} > \mu_t$, so the standard multiplicative Poisson-binomial Chernoff bound gives

$$\mathbb{P}[Y_t \geq n] = \mathbb{P}[X_t \geq n-1] \leq \left(\frac{e\mu_t}{n-1} \right)^{n-1} \leq \alpha_n.$$

Since the stopped real chain and Y have the same first hit of n ,

$$\mathbb{P}[\tau_n \leq m] \leq \sum_{t=0}^m \mathbb{P}[Y_t \geq n] \leq (m+1)\alpha_n.$$

For $n \geq 14$,

$$\alpha_n \leq \alpha_{14} < \frac{1}{16}.$$

Indeed $e\sqrt{n}/(n-1)$ is decreasing in n and is already below 1 for $n \geq 10$; therefore α_n is decreasing from that point onward. Take

$$m = \left\lfloor \frac{1}{8\alpha_n} \right\rfloor.$$

Then

$$(m+1)\alpha_n \leq \frac{1}{8} + \alpha_n < \frac{1}{2},$$

so

$$\mathbb{P}[\tau_n > m] > \frac{1}{2}.$$

Moreover $\alpha_n < 1/16$ implies $m \geq 1/(16\alpha_n)$. Hence

$$\mathbb{E}[\tau_n] \geq m \mathbb{P}[\tau_n > m] \geq \frac{1}{32\alpha_n} = \Omega(\alpha_n^{-1}).$$

Finally,

$$\log_2 \alpha_n^{-1} = (n-1) \left(\log_2(n-1) - \frac{1}{2} \log_2 n - \log_2 e \right) = \frac{1}{2}n \log_2 n - O(n).$$

□

Theorem 9 (Fixed-bias BRF-family minimax order). *Let $p_n \in (0, 1)$ be any size-dependent choice of homogeneous BRF survival bias, and let*

$$W_n(p_n)$$

be the causal state-adaptive worst-case expected stabilization time. Then

$$W_n(p_n) = 2^{\Omega(n \log n)}$$

uniformly over all such bias sequences. Together with the dyadic size-aware upper bound,

$$\inf_{p \in (0,1)} W_n(p) = 2^{\Theta(n \log n)}.$$

Proof. Write $\delta_n = 1 - p_n$.

If

$$\delta_n \leq n^{-1/2},$$

the universal simultaneous-drop lower bound gives

$$W_n(p_n) \geq \delta_n^{-n} \geq n^{n/2} = 2^{\frac{1}{2}n \log_2 n}.$$

If

$$\delta_n > n^{-1/2},$$

choose the admissible one-frontier adversary. Stabilization cannot occur before the active set has hit size n , so

$$W_n(p_n) \geq \mathbb{E}[\tau_n].$$

For $n \geq 14$, the preceding lemma gives

$$W_n(p_n) \geq 2^{\frac{1}{2}n \log_2 n - O(n)}.$$

Thus every homogeneous fixed-bias BRF choice has $2^{\Omega(n \log n)}$ state-adaptive worst-case expected time.

The size-aware dyadic choice from the first theorem satisfies

$$W_n(p_n) \leq 2^{n \log_2 n + O(n)}.$$

Combining the two bounds proves the minimax order. $\square$

Remark 10 (Scope of the optimality claim). *The previous theorem is only an optimality result over homogeneous fixed-bias BRF protocols: one survival probability is used by every active node in every round for the given network size n . It does not exclude faster size-dependent protocols with additional control structure, time-varying probabilities, history-dependent probabilities, or different two-state randomized transition rules.*

7 Computational evidence

The theory above is independent of computation. We nevertheless include finite-size Bellman calculations for three reasons: to regression-test the general- p equations against the predecessor's exact fair-coin formula, to show the practical size of the knowledge-induced speedup at moderate n , and to compare the proof-oriented dyadic bias with biases suggested by the exact finite-state Bellman system.

7.1 Method

By Theorem 6, one-frontier exposure is worst-case for every fixed bias p . Therefore finite worst-case expectations can be computed without enumerating graph sequences. For states $k < n$ we solve

$$E_k = 1 + \sum_{j=0}^k \binom{k}{j} p^j (1-p)^{k-j} E_{j+1},$$

and at the all-active state

$$E_n = 1 + \sum_{j=1}^n \binom{n}{j} p^j (1-p)^{n-j} E_j, \quad E_0 = 0.$$

For rational p , the accompanying script solves this $n \times n$ linear system with exact rational Gaussian elimination. Fair, dyadic, and certificate-optimal biases are therefore evaluated exactly for $2 \leq n \leq 20$. For the finite-size runtime optimum, the same Bellman system is solved at 90 decimal digits while a one-dimensional golden-section search minimizes $\log_2 E_1$ over $p \in [0.05, 0.9999]$. The numerical optimum is supporting evidence only and is not used in any theorem.

As a regression gate, exact rational evaluation at $p = 1/2$ reproduces the predecessor values

$$10, 58, 602, 14106, 755610, 87665306, 21246839962$$

for $n = 2, \dots, 8$. The same calculation extends them to

$$E_1(9) = 10550999832730, \quad E_1(10) = 10622132397727898.$$

7.2 Finite-size comparison

Table 1 compares the fair, dyadic, certificate-optimal, and numerically located runtime minima. Values are reported on a base-two logarithmic scale because the expectations become enormous even at moderate n .

n	p_{dy}	p^*	p_{num}	fair	dyadic $\log_2 W_n(p)$	cert.	num.
4	0.7500	0.6000	0.5933	9.23	9.89	8.94	8.94
6	0.7500	0.7143	0.7157	19.53	16.08	15.95	15.95
8	0.8750	0.7778	0.7827	34.31	26.35	23.85	23.85
10	0.8750	0.8182	0.8246	53.24	33.75	32.44	32.42
12	0.8750	0.8462	0.8531	76.22	41.93	41.58	41.54
16	0.9375	0.8824	0.8893	134.21	66.83	61.19	61.12
20	0.9375	0.9048	0.9112	208.21	85.04	82.23	82.11

Table 1: Finite Bellman comparison. $p^* = (n-1)/(n+1)$ maximizes the block-success certificate, while p_{num} is the high-precision minimum located by one-dimensional search on the exact Bellman expectation.

The main pattern is robust. Fair BRF rapidly separates from all size-aware choices. For example, at $n = 10$ the exact Bellman values give $\log_2 W_{10}(1/2) \approx 53.24$, versus 33.75 for the dyadic bias and 32.44 for the certificate-optimal bias. The numerically located value is 32.42. Thus the size-aware improvement is already large at modest n , while the certificate-optimal continuous bias is close to the numerical runtime optimum.

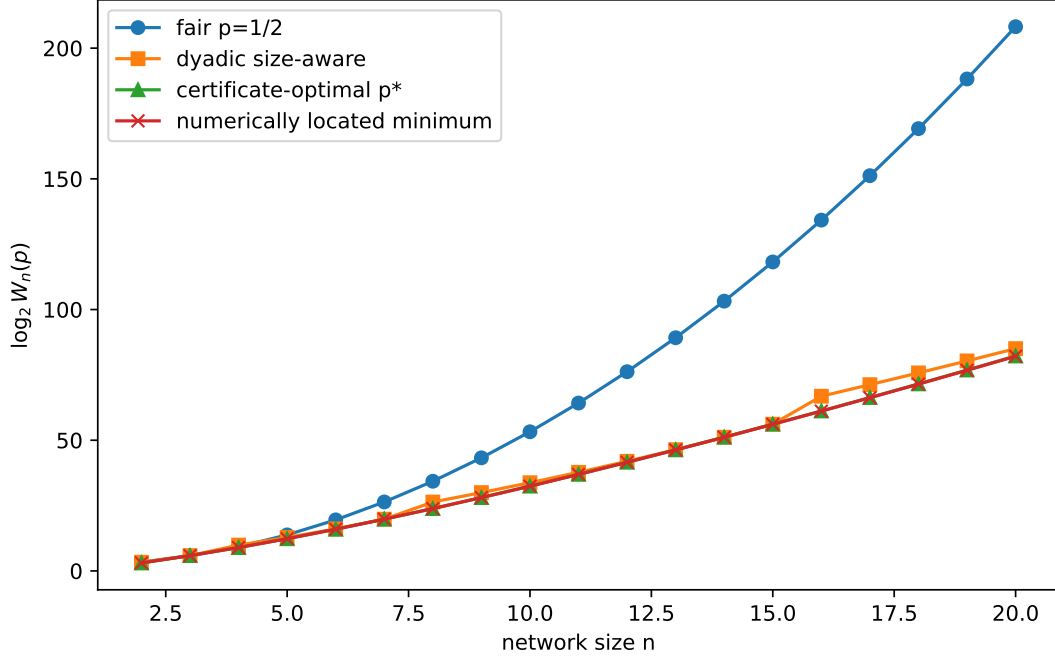


Figure 1: Finite causal state-adaptive worst-case Bellman values. The fair-coin curve grows on the n^2 exponent scale, while size-aware choices lie on the much smaller $n \log n$ scale. The numerical curve is descriptive, not a theorem.

7.3 Bias landscape

Figure 2 shows a sampled bias landscape at $n = 10$. The certificate maximizer $p^* = 9/11 \approx 0.81818$ lies close to the numerically located runtime minimum $p_{\text{num}} \approx 0.82458$. The dyadic choice $p_{10} = 7/8 = 0.875$ is deliberately constrained to a fair-coin-friendly probability and is slower at this finite size, although it has the same proved asymptotic order.

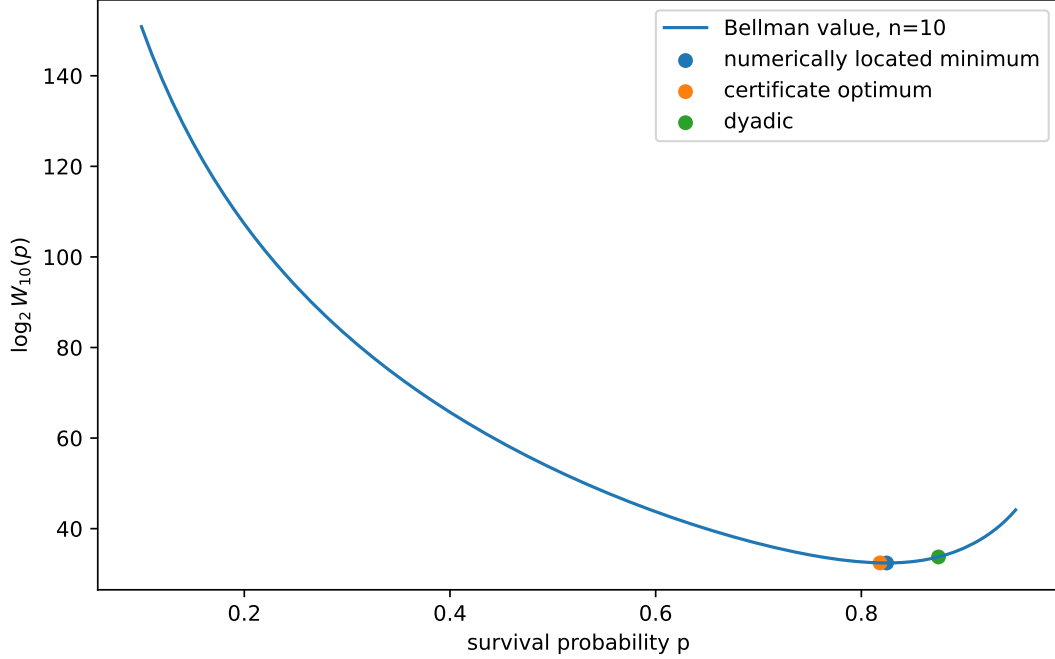


Figure 2: High-precision Bellman bias landscape for $n = 10$. Markers show the numerically located runtime minimum, the block-certificate maximizer, and the dyadic size-aware choice.

Across $2 \leq n \leq 20$, the certificate-optimal bias remains close to the numerically located minimum: their difference in $\log_2 W_n$ is below 0.121 bits in the audited range. This observation does not prove that p^* is asymptotically or exactly runtime-optimal; it only motivates the sharper open question of characterizing the true finite- n optimal fixed bias.

8 Scope, limitations, and open problems

The safe headline claims supported by the proofs are:

1. known-size dyadic BRF retains exactly two persistent protocol states and achieves $2^{\Theta(n \log n)}$ causal state-adaptive worst-case expected stabilization time;
2. for arbitrary fixed $p \in (0, 1)$, one-frontier exposure is Bellman-optimal for homogeneous BRF; and
3. $2^{\Theta(n \log n)}$ is the minimax order over homogeneous fixed-bias BRF.

The paper does *not* claim that one persistent bit is the randomized minimum, that $2^{\Theta(n \log n)}$ is optimal over all two-state randomized broadcast protocols, or that either the dyadic bias or p^* exactly minimizes the actual expected stabilization time. Time-varying biases, history-dependent randomization, and richer two-state transition kernels remain outside the lower bound. The tight worst-case behavior against oblivious topology adversaries also remains open, as does the stronger termination-detection boundary.

9 Conclusion

Network-size knowledge can buy a large running-time improvement without buying additional persistent protocol state. In BRF, fair private randomness already permits idle-start stabilizing broadcast with two persistent states, but a size-oblivious fair bias pays $2^{\Theta(n^2)}$ adaptive worst-case time. Compiling the known size into the survival probability reduces this to $2^{\Theta(n \log n)}$, and no homogeneous fixed bias can improve that order against a causal state-adaptive adversary. The resulting picture separates three resources that are often conflated: persistent local state, externally supplied problem knowledge, and time.

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